

type of toughened glass. Silicon solar modules have been in the field for more than 50 years and perform quite predictably. These are guaranteed for 25 years of field life but the power yield drops about 0.6 per cent a year. One can use monocrystalline (made from a single crystal) or polycrystalline (made from multiple crystals) panels. Monocrystalline panels are a little more efficient but the cost per watt is almost the same.

Copper-indium gallium-diselenide (CIGS) panels may become the preferred type for commercial rooftop projects in another five years. These have the potential to deliver reasonable efficiencies at a lower cost than traditional crystalline panels. However, the cost per watt may not necessarily go down, only the panel size per watt may drop. Today, solar panels (depending on the brand) are bankable, that is, banks loan capital for their purchase.

The solar energy can be used for captive consumption or exported to the grid. The electrical energy (DC) or the solar power generated by the solar PV modules during the sunshine hours is stored in the batteries for use, as and when required. The energy stored in the batteries is converted into 230V AC mains using an inverter. This energy automatically synchronises with the grid and gets injected into it.

Installation by integrators

Many solar system installers and owners have had good experience in anchoring the panel structures. This has to be done scientifically and with care. It is possible to have non-anchored installation systems but these need to be very carefully designed to withstand heavy winds. Such systems are designed to connect the solar power system to a roof using weights, rather than fasteners that must be anchored to the roof.

Solar installation companies, often called integrators, can complete a small rooftop project within a few weeks. Before signing a contract with an

How to supply solar power to the grid?

If the solar power generated from a rooftop installation is to be injected into the grid, one needs to enter into a power purchase agreement (PPA) with the local distribution utility in whose area the solar system is located. Under this agreement, a tariff is determined by the appropriate State Electricity Regulatory Commission (SERC). However, the issues related to grid integration, metering, measurement and energy accounting for projects are under consideration with the government.

There is no cost involved in the transmission of energy unless the power is transmitted at high tension (HT) levels (11 kV or 33 kV), and special monitoring and metering hardware are deployed at HT levels. In the current scenario, metering arrangements for rooftop grid-interactive power plants are under active consideration by the government.

While no special arrangements are required to inject power into the grid, there is a safety aspect that needs to be factored in while transmitting energy. There is always a risk involved, as when the grid fails the solar power system automatically stops injecting power into the grid. This is called islanding, where the inverter isolates itself. This is a standard feature built into solar power inverters, making these safe for residential and commercial applications. A standalone feature in the inverter would enable captive consumption of the solar power generated in the event of any grid outage.

integrator, evaluate the roof for solar installation with respect to:

Roof condition. The roof should be in a good state prior to solar installation. If it needs significant repair or replacement, get this done before installing the solar array.

Space availability. Solar power projects work best on flat roofs without obstructions.

Weight load. Some roofs are not designed to hold much additional weight. Ascertain the acceptable weight you can add to your roof before signing a contract.

Investments involved

Of all the components of a solar PV plant, solar module accounts for the biggest cost—it can be 70 per cent of the total project cost. The cost per watt is currently Rs 130-150 ex-factory. The investment primarily depends upon the size of the power plant, which varies from a small kilowatt to multi-megawatt plant. At present, good-quality off-grid rooftop solar power plants can be installed at a cost of Rs 250,000 per kW.

Under the National Solar Mission policy, the benchmark price for an off-grid system is Rs 270,000 per kW peak. For a grid-connected system, it is Rs 190,000 per kW peak. The government also provides a 30 per cent subsidy on the benchmark price. Installation costs

would differ in case of remote installations and poor site conditions.

Economic advantages

The total investment per kW in a small power plant, for example, 10kW, will be the same if not less than in a large 5MW plant. It is therefore viable to go for small grid-fed plants owned by small privately-owned utilities. A buying rate of Rs 17 or 18 per unit of electricity from such plants will attract thousands of small investors like a magnet. In most areas in India, solar power can then be a fiscally sound investment that reduces electricity payments immediately, as well as hedges the small solar plant owners against local utility price increases.

If solar power is fed into a small city grid like Miraj, Ratnagiri or Ratlam, all consumers in that area will get cleaner uninterrupted power from the local copper grid. The high impedance of the local grid helps power to remain local, improving the local power quality. That's why all other countries in the world allow solar plant owners to feed power into the local grid at the low voltage end.

It therefore makes sense to set up a solar rooftop plant in cities or towns facing severe electricity shortages. Today, the cost of generating electricity using a diesel generation (DG) set is in the range of Rs 20-22 per unit, whereas